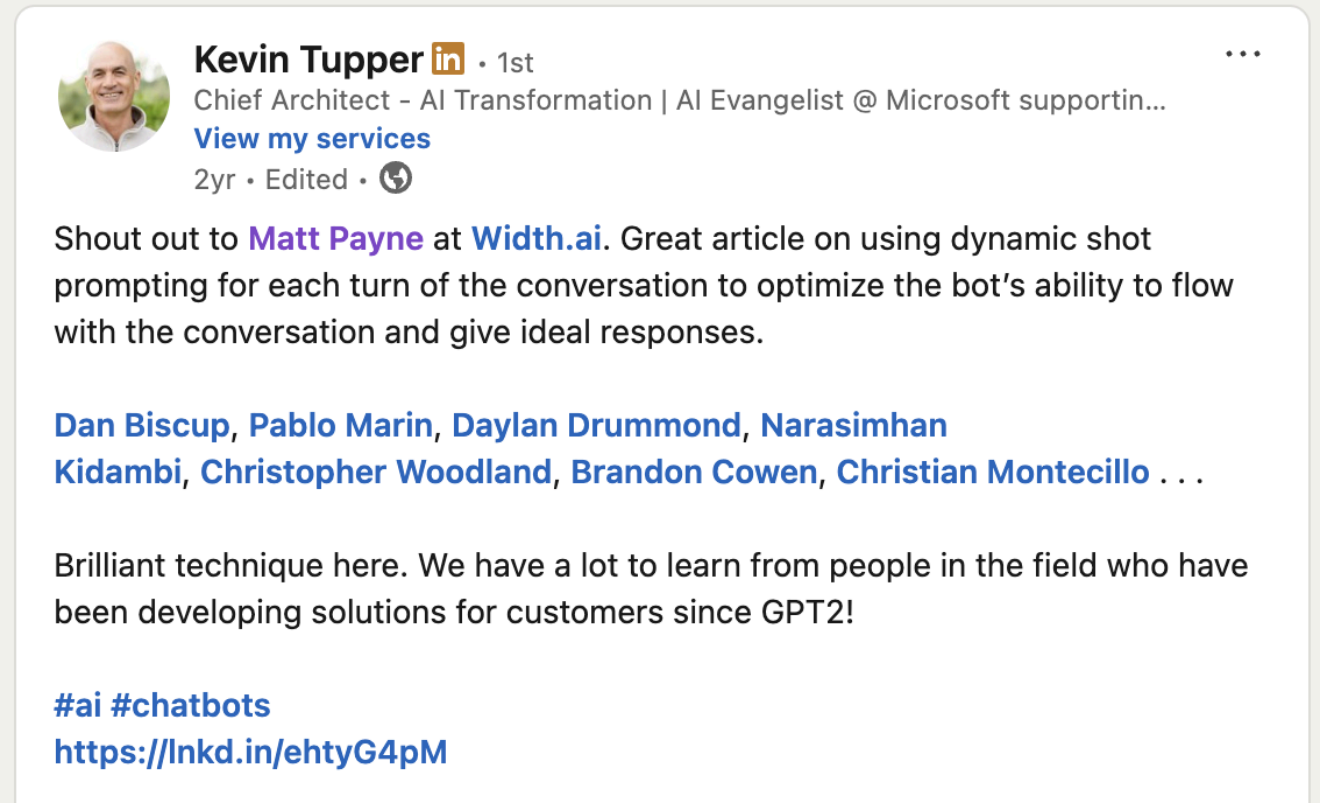
Width.ai

Custom AI & Machine Learning Development

Document Processing — Work Summary

Production document-processing systems Width.ai has built across medical, legal, financial, e-commerce, HR, education-technology, and enterprise domains — organized by the methods and technical approaches we use, with measured results.

What they say



Kevin Tupper • 1st

Chief Architect - AI Transformation | AI Evangelist @ Microsoft supportin...

View my services

2yr • Edited •

Shout out to Matt Payne at Width.ai. Great article on using dynamic shot prompting for each turn of the conversation to optimize the bot's ability to flow with the conversation and give ideal responses.

Dan Biscup, Pablo Marin, Daylan Drummond, Narasimhan Kidambi, Christopher Woodland, Brandon Cowen, Christian Montecillo . . .

Brilliant technique here. We have a lot to learn from people in the field who have been developing solutions for customers since GPT2!

#ai #chatbots

https://lnkd.in/ehtyG4pM

Ai Engine Dev

5.0 | Jul 28, 2025 - Sep 26, 2025

"Working with Matt on the development of our AI engine has been an outstanding experience from start to finish. His deep expertise with LangChain and OpenAI was evident throughout the project, as he not only guided the architecture but also ensured seamless integration with our existing systems.

What sets Matt apart is not just his technical skill, but his ability to clearly explain complex concepts and translate them into practical solutions that drive results. He and his team demonstrated exceptional problem-solving abilities, a forward-thinking approach, and a genuine commitment to excellence.

The professionalism and collaboration from his team were second to none. They were highly responsive, innovative in their ideas, and consistently delivered above expectations. Every milestone was achieved on time, and the final AI engine exceeded our goals for performance, scalability, and usability.

If you are looking for a partner who combines deep technical expertise with an exceptional team culture, I cannot recommend Matt and his team highly enough. They truly set the standard for excellence in AI development." less



Matt is an AI expert in a sea of could/would/should be experts. He is a quick learner, has a sharp mind for business, and knows how to bring AI to organizations struggling to make sense of it all. We used him as a consultant and will re-engage as our internal discussions and project planning progress.

Nick Matteucci
CEO, WorkOtter



Matt is an expert. He's also a friendly, flexible, and able communicator. He came through just as promised and was able to suggest a helpful work flow for the project. I hope and plan to work with him again.

Jim Dennen
Professor, Denison University



Matt is an experienced AI expert who can shortcut your time to market. He has a wealth of knowledge and is very willing to share it.

Mic Black
Mic's Lab Pty Ltd

Overview

Width.ai builds custom document processing systems that consistently outperform off-the-shelf LLMs and generic OCR/RAG products on real client data. Our approach combines fine-tuned document-specific transformer architectures (Donut, DETR, LayoutLMv3, fine-tuned LLMs), custom OCR and layout-understanding pipelines, program synthesis for structured output, agentic retrieval for multi-document Q&A, and multi-knowledge-base context management. Every system is built for the specific domain, document types, and downstream use case of the customer.

The work below has been deployed across medical record summarization, legal cover sheet processing, invoice line-item extraction, deposition transcript Q&A, contract clause rewriting, marketplace product categorization, resume parsing, and content moderation — with multiple engagements directly enabling 8 figure Series A fundraises for our customers.

Featured Case Studies

Patient Record Summarization — Medical AI Startup

Built a production-level medical document summarization system that ingests varied patient record layouts — including OCR-processed scans — and reliably extracts essential medical data: names of medical personnel, diagnoses, ICD codes, medications, and dates of visits and procedures. All extracted data rolls into a single patient timeline summary, plus document-level summaries per record. Now deployed in production across both medical and legal industries with a customizable pipeline at each step. Two separate healthcare companies used this work to raise rounds.

- **96%+ accuracy on the top 10 extracted fields.**
- **90%+ accuracy on every extracted field.**
- **Customizable pipeline at each step.**
- **Fine-tuned model served via Ollama, with a GPT-backed option for the same pipeline.**
- **The startup raised its Series A on the back of this work.**
- **Full architecture: med-document-processing**

Case studies: patient-record-summarization · gpt-4-medical-record-summarization-pipeline

Patient Records Q&A System

Q&A system built on the same medical document architecture. The pipeline uses LayoutLMv3 plus a program-synthesis layer to classify page type and visit context, so questions about referring physicians, home-health-care providers, and dosage information return answers attributed to the correct document and visit — not just the first match on the page. A medical record-matching architecture links related documents across the same patient even when patient information varies between records.

- 50 preset clinical questions all answered at >91% accuracy.
- Plus open-domain handling for questions outside the preset set.
- LayoutLMv3 + program synthesis for page-type and visit-context classification.
- Medical record matching architecture for cross-document linking.

Architecture docs: LayoutLMv3 + program synthesis · medical record matching

CaseUp.ai — Deposition Transcript Analysis

Multi-deposition transcript Q&A and conversational system handling search up to 10,000 pages with multi-step and multi-topic queries. Built on a planning + agentic retrieval framework where agents leverage multiple agent/tool pairings to search documents.

- Search up to 10,000 pages per session.
- Multi-step and multi-topic queries supported.

- Search tools: grep/glob, metadata via generated metadata and abstractive fields, query augmentation.
- System supports retries and validation at the agent level.

PDF Transcript Processing

During pdf processing, we use a custom claude prompt system for per page processing to tie line numbers to specific statements. During this step we also generate per topic metadata that uses a sliding window to pull all transcript content associated with a specific topic. This allows us to create per page metadata that is not limited to a specific chunk or page, but a dynamic system to provide deeper context. All LLM prompt driven.

Invoice Document Processing — Fine-Tuned Donut & Custom Paddle Architecture

Fine-tuned the Donut (Document Understanding Transformer) architecture + Paddle OCR architecture for invoice processing, restructuring it to organize data variance in column-to-line-item relationships. Implemented program synthesis architecture to tie keys/value pairs to top-level columns on a fine-tuned relationship.

- 95.6% accuracy.
- Outperformed off-the-shelf LLMs at the time.
- Program synthesis architecture for key/value-to-column relationships.
- Adjusted weights of Paddle OCR to focus on text size similarities in OCR.

Architecture docs: invoice processing

Legal Cover Sheets at Scale — Fine-Tuned DETR

Fine-tuned DETR document transformer architecture for processing legal document cover sheets at scale across all federal and state-specific formats. Production system handles real-world degradation modes that off-the-shelf document parsers fail on.

- 96.5% accuracy across all federal and state formats.
- Handwriting processing.
- Checkbox recognition.
- Low-quality scan handling.

System has since moved to an LLM driven pipeline with OCR confidence scoring. Best of both worlds.

Case study: https://www.width.ai/post/information-extraction-legal-documents

Resume Parser / Document Processor

Production document processor for resumes — a domain with no generic templates and high format variance. Reached state-of-the-art accuracy outperforming general-purpose LLMs on the same task.

Achieved 94% accuracy across all fields and all resume formats. Our accuracy on the most common 1 and 2 column resumes reached 98% across all fields.

- 1.7s processing on just 4 cores.
- Outperforms ChatGPT on the same task.

1 and 2 Column Resumes

The accuracy with the most common formats was ridiculous. 98% across all fields in a very large sample size. By the end of fine-tuning popular fields such as phone number, skills sections, and email reached 99%+. The overall accuracy when including all 1 and 2 resume formats was still over 94%.

3 and 4 Column Resumes

The contact information related fields in these noisy resumes had an accuracy of 93.2% across the fields.

Case study: width.ai/post/resume-parser-software

Legal Clause Rewriter — Legal-Tech Startup

NLP system that rewrites contract clauses to improve clarity and reduce downstream legal risk. The customer raised its Series A on the back of this work.

Case study: building-an-nlp-legal-clause-rewriter

PEARL — Long-Form Document Summarization Framework

Implemented a state-of-the-art prompting framework for executing planned actions across very long, domain-specific documents. Enables abstractive summarization at lengths and accuracy levels traditional methods can't reach.

Case study: prompting-large-language-models-to-plan-and-execute-actions-over-long-documents-pearl

Multi-Knowledge-Base Document Q&A — NILES

Built from scratch the ability for users to upload their own documents into an enterprise conversational product. The context-management system juggles the platform's core knowledge base, organization-specific knowledge bases, and user-provided documents — answers understand this hierarchy and prioritize information correctly.

askniles.ai Platform has 10k plus customers and multiple enterprise level organizations using conversational RAG every day. System supports custom knowledgebases for each organization and the retrieval system handles the data variance that comes with it.

Built user uploaded document processing system to handle on the fly any pdf, word document, or markdown document. Uses LLM prompt system to generalize the processing for chat.

Methods & Technical Approach

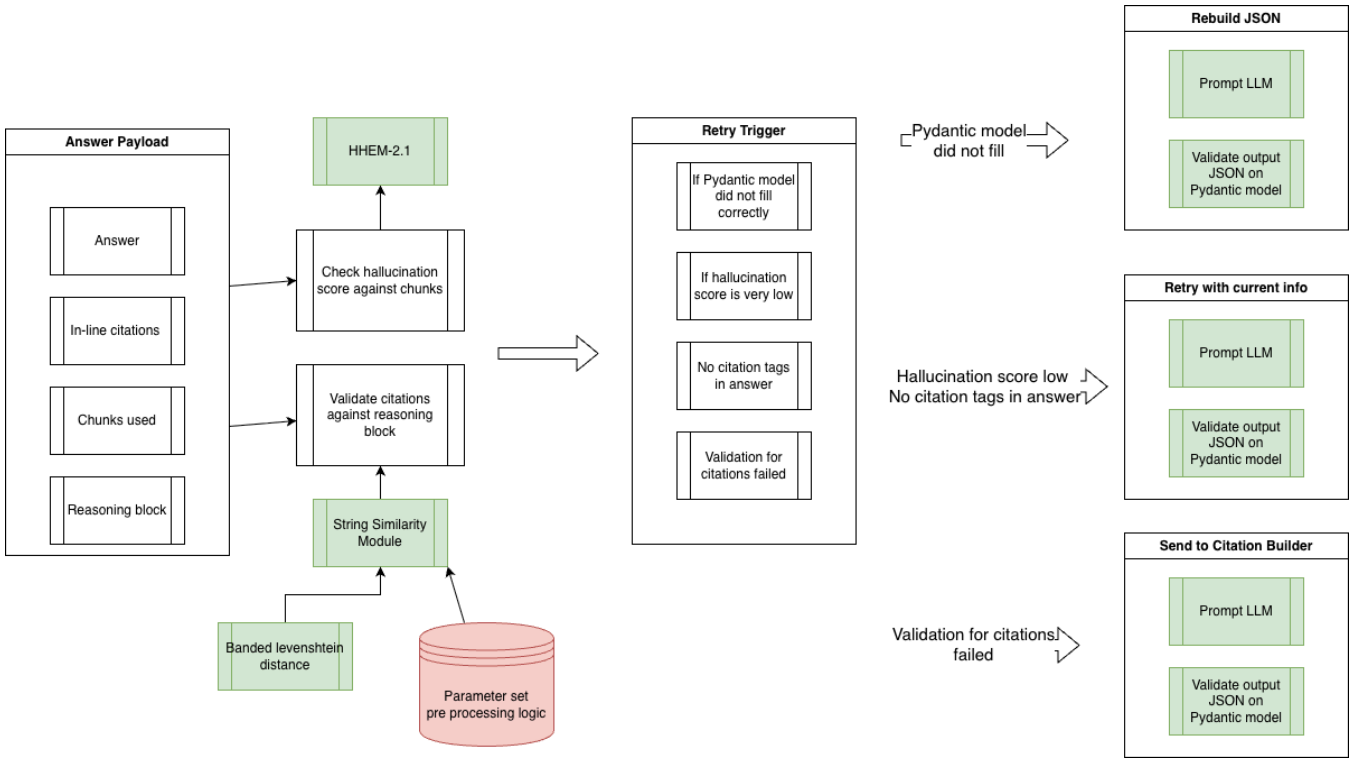
Hierarchical Relationship Extraction using Program Synthesis

Our SOTA program synthesis system for document processing allows us to build a system that uses machine learning to learn key -> value attribution rules (entity linking) through a trained relationship on our most popular documents. Gives us an understanding of key -> value distance, entity mapping with boxes and tables, and character size relationships. Trained LayoutLM v3 across common layouts.

Based on this research: arxiv.org/pdf/2106.00980

LLM Output Validation for document processing

Simple method to validate output json against specific checks from document processing and RAG LLM use cases. Helps validate the output and perform automatic retries to fill data. Generates metrics as well.



ReAct Prompting - Built in Validation & Recursion

Prompting framework used in many agentic systems as the main framework for agents to use. Very useful on document processing use cases to leverage built in validation of fields and recursion.

- **Link: https://www.width.ai/post/react-prompting**
- **Talk I gave on ReAct at a conference in 2024: https://docs.google.com/presentation/d/1-kEsjlbZemrx8uvq-l-doYNAoQIEJFQFtXSfSMWIeNM/edit?usp=sharing**

Layout Schema Driven Prompt Tuning

Simple but very effective technique that adjusts the prompt rules for the LLM document processing step based on a layout understanding model. This can be used to do things like:

- **Layout prompt rules —** Based on recognized formats we can adjust the rules given to the LLM to help it understand focus points. Our layout understanding model identifies a table, which we know based on the document classification is a specific page type. Rules are applied specific to that page type
- **Field Focused rules —** Based on recognized entities from the layout understanding model, we can provide dynamic rules to the LLM giving a bit of guidance on how to use the information. On an invoice we recognize a “total_cost” field and rows and columns. Rules can be created to help the LLM understand how to tie the two together in the final JSON.
- **Quick data recognition—** Helps with hallucinations. Instead of providing the LLM every time with the rules for data extraction, even on a page that is completely irrelevant, keeps it from hallucinations. If we know in the full document some pages are not useful for extraction, and that better pages exist for these data points, we can provide a different ruleset for this irrelevant page, or skip altogether.

Can use things like Surya, TATR, or LayoutLM.

Fine-Tuned Document Transformer Architectures

We fine-tune document-specific transformer architectures (Donut, DETR, and fine-tuned open-source LLMs like Qwen 2.5) rather than relying on general-purpose LLMs for extraction. This delivers higher accuracy on domain-specific tasks at lower inference cost.

- **Donut (Document Understanding Transformer) —** Restructured the Donut architecture for invoice line-item extraction to organize data variance in column-to-line-item relationships. 95.6% accuracy, outperforming off-the-shelf LLMs at the time.
- **DETR (Detection Transformer) —** Fine-tuned for processing legal document cover sheets at scale across all federal and state-specific formats. 96.5% accuracy across all formats. Handles handwriting processing, checkbox recognition, and low-quality scans.
- **Fine-tuned Qwen 2.5 —** Privately-instanced content-moderation agent for an education-technology platform. 100% accuracy on the moderation filter in blind evaluations; <1% hallucination rate on domain-specific tasks via end-to-end fine-tuning with synthetic data generation, reinforcement learning, and blind evaluations.

Keys

Hallucination score checking

We used a fine-tuned HHEM-2.1 hallucination detection model. It looks at the answer text, looks at the chunks used, and produces a single score.

String Similarity

We used a custom built banded levenshtein distance function that allowed for domain specific tuning. Was much faster than out of the box LS distance.

Pydantic model

We used a pydantic model to ensure the structured output with the required fields we need. If the answer does not have valid JSON that fills the Pydantic model, we move to retry logic that first tries a simple rebuild approach with key/value pairs, and then just uses a rebuild prompt.

Citations

If the citations are missing completely we retry with the entire input. No point in rebuilding if we don't have any.

If the validation of citations fails we use a citation builder that takes in our info and tries to create new citations.

<cite>NasscoInc - 45:19-46:3</cite>

<cite>NasscoInc - 45:19-45:20</cite>

This is a key add as using the full chunk as a citation is poor tracing.

OCR & Layout-Understanding Pipelines

Custom OCR pipelines built and tuned to feed downstream NLP and extraction systems with reliable, high-fidelity text — even on degraded inputs. Layout-aware extraction across heterogeneous document formats, including table detection, structure recognition, and functional analysis.

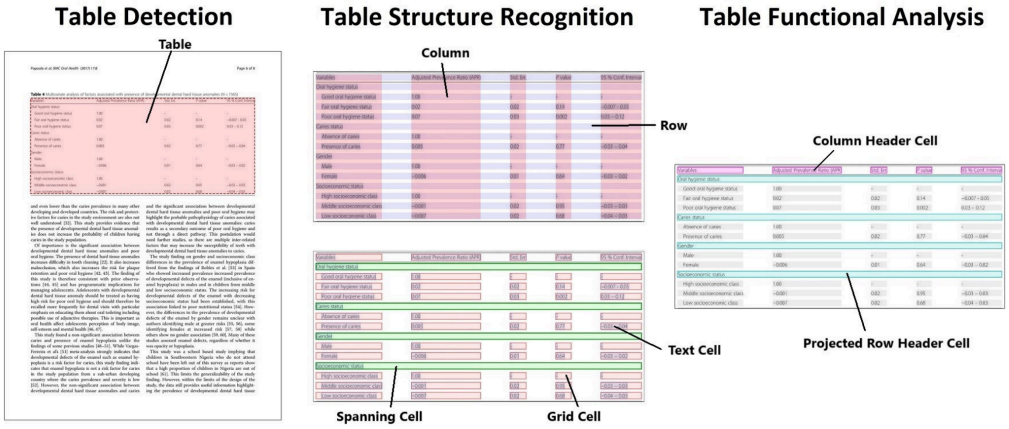


Table detection, structure recognition, and functional analysis stages in our layout-understanding pipeline.

- Handwriting, checkbox, and low-quality-scan handling for legal forms.
- OCR pipelines feeding downstream NLP at production volumes.
- Layout-aware extraction across heterogeneous formats with no generic templates.

Layout-Aware Models + Program Synthesis

When document data must conform to structured schemas — line items in invoices, fields in patient records, columns in tables — we implement program synthesis architectures that tie key/value pairs to top-level columns on a fine-tuned relationship. Paired with layout-aware visual document models like LayoutLMv3, this handles the data variance that plain LLM extraction and pure OCR both fail on.

- **LayoutLMv3 —** Visual document model for page-type understanding and structured field construction. Integrated with program synthesis for patient records Q&A so the system knows which page type it's reading before answering.
- **Program synthesis architecture —** Used for invoice line-items (tying key/value pairs to top-level columns on a fine-tuned relationship) and for medical record matching across documents with varying patient information.

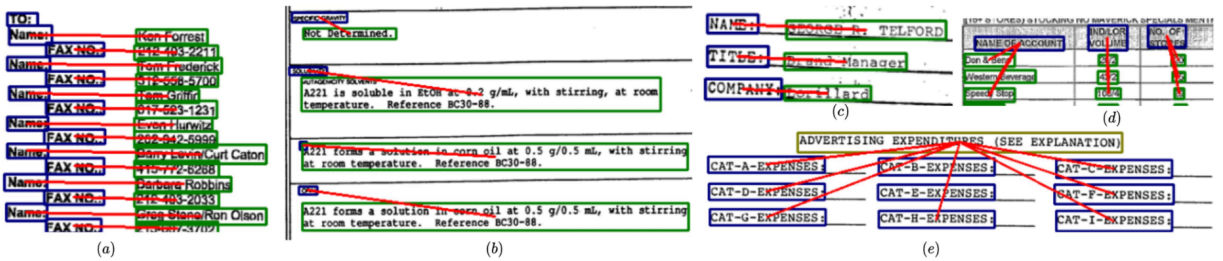


Figure 1: Example output of VRDSynth's synthesized programs in different layout: (a) List of key-value group, (b) key and long value, (c) list of key-value (flat) (d) table and (e) header and key-value table. The programs collectively link each corresponding key (e.g., names, fax) to corresponding values or headers to corresponding keys. Note that we show the original images to also demonstrate the problem of varying quality of scanned VRDs.

Agentic Retrieval for Document Q&A

For multi-document Q&A at scale, we build agentic frameworks rather than single-shot RAG. Agents plan over the document set, invoke multiple search tools, and validate before answering — enabling multi-step and multi-topic queries that single-embedding retrieval cannot serve. Search tools available to the agent include:

- grep / glob over document text.
- Metadata search via generated metadata and abstractive fields.
- Query augmentation.
- Hybrid semantic + keyword search (where the domain requires both, e.g. product names that pure semantic search misses).

Frameworks support agent-level retries and validation, with deterministic and qualitative output validation on every step.